

THE REAL-ROOTEDNESS PROBABILITY OF A RANDOM CUBIC POLYNOMIAL: THREE CLOSED FORMS

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ABSTRACT. We compute, in closed form, the probability that a random cubic polynomial has three real roots, in three related models. For a *monic* cubic $x^3 + ax^2 + bx + c$ with (a, b, c) independent and uniform on $[-1, 1]$, this probability is $\frac{383}{4860} + \frac{\ln 3}{48}$; on $[0, 1]$ it is exactly $\frac{1}{2880}$. For the *non-monic* cubic $ax^3 + bx^2 + cx + d$ with all four coefficients independent and uniform on $[-1, 1]$, it is $\frac{641}{2430} - \frac{\ln 3}{24}$. The first two values appear to be new. The third confirms, with a first proof, a closed form that had circulated unproved in an informal online forum thread since June 2026. All three results are verified by independent symbolic and numerical computation, and by a complete, zero-~~sorry~~ machine-checked proof of all three theorems in Lean 4 / Mathlib – which along the way proves, from scratch, the classical fact connecting a real cubic’s discriminant sign to its real-root count, seemingly absent from Mathlib until now. We record the reduction technique for each case – an elementary band/discriminant-sign argument for the monic case, and a divergence-theorem argument exploiting the homogeneity of the discriminant for the non-monic case, which formalization showed can be replaced by a strictly more elementary slicing argument – since both appear to generalize to related open problems, which we list.

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With assistance from Claude (Anthropic). Machinery, proofs, and their verification (symbolic, numerical, and partially machine-checked in Lean 4 / Mathlib) are recorded in the accompanying repository.

1. INTRODUCTION

How likely is a “random” polynomial to have all of its roots real? For degree 2 this is classical. For a monic quadratic $x^2 + bx + c$ with (b, c) uniform on $[-1, 1]^2$, real roots occur exactly when $c \leq b^2/4$, giving probability $13/24$; for the full quadratic $ax^2 + bx + c$ the answer involves a logarithm, $\frac{41}{72} + \frac{\ln 2}{12}$ on $[-1, 1]^3$ and $\frac{5}{36} + \frac{\ln 2}{6}$ on $[0, 1]^3$, the latter apparently first recorded (without derivation) by J. D’Aurizio and rederived in [3]. For the analogous *cubic* question the literature is far thinner. The only paper we could locate that claims exact results for random cubics with general coefficient distributions is Li [1], which is effectively unobtainable (two citations in thirty-eight years, neither reproducing a value) and, per the recovered *Mathematical Reviews* notice [2], treats a family of symmetric-interval distributions that does not include the one-sided case of our Theorem 2 at all, and reaches our Theorem 1 only as an unverifiable special case.

This note settles three instances of the cubic question completely.

Theorem 1 (Monic cubic, symmetric cube). *Let (a, b, c) be independent and uniform on $[-1, 1]$. Then*

$$\mathbb{P}(x^3 + ax^2 + bx + c \text{ has three real roots}) = \frac{383}{4860} + \frac{\ln 3}{48} = 0.10169434037605886959954086 \dots$$

Theorem 2 (Monic cubic, unit cube). *Let (a, b, c) be independent and uniform on $[0, 1]$. Then*

$$\mathbb{P}(x^3 + ax^2 + bx + c \text{ has three real roots}) = \frac{1}{2880}.$$

Theorem 3 (Non-monic cubic). *Let (a, b, c, d) be independent and uniform on $[-1, 1]$. Then*

$$\mathbb{P}(ax^3 + bx^2 + cx + d \text{ has three real roots}) = \frac{641}{2430} - \frac{\ln 3}{24} = 0.21801049620261477108898412335868 \dots$$

Theorems 1 and 2 are, to our knowledge, new: an eight-agent literature sweep together with a targeted search for the exact constants (Section 6) found no prior publication of either value, digit string, or exact expression. Theorem 3 is more subtle. The exact value $\frac{641}{2430} - \frac{\ln 3}{24}$ already existed, unproved, as the terminus of a long, informally written, partly AI-assisted derivation posted to a Russian mathematics forum (dxdy.ru) in June 2026. Section 4 gives the first proof of this value, together with five independent lines of numerical and symbolic corroboration.

Motivation and context. The cubic question is the tractable sibling of a harder open problem: for an $n \times n$ matrix with i.i.d. uniform entries, what is the probability that all n eigenvalues are real? For $n = 2$ this is classical ($49/72$ for entries uniform on $[-1, 1]$, [4]), but already for $n = 3$ no closed form or rigorous nontrivial bound is known; the canonical statement of the question, posted to Mathematics Stack Exchange in 2020, has 153 upvotes and, as of this writing, zero answers [6]. The obstruction is structural: for Gaussian entries, orthogonal invariance gives Edelman’s exact $2^{-n(n-1)/4}$ [5], but the uniform cube has no such symmetry to exploit. Moving from matrix-entry space to *coefficient* space – asking instead for the probability that a random *characteristic polynomial* is real-rooted – keeps the same underlying object (the volume of a semialgebraic region cut out by a discriminant) while dropping the ambient dimension from 9 (a 3×3 matrix) to 3–4 (a cubic’s coefficients), where it becomes tractable by elementary means. This note is the result of attacking that lower-dimensional sibling.

Machine verification. Every calculus step in the proofs of Theorems 1–3 is checked twice over in exact arithmetic (once by hand below, once independently by computer algebra), confirmed by high-precision quadrature and by direct Monte Carlo sampling, and – new for a paper at this level of the literature – verified by a *complete, zero-sorry machine-checked proof of all three theorems* in Lean 4 against Mathlib, described in Section 5. That formalization also produced a result of independent interest: a from-scratch, calculus-free proof that a real cubic’s discriminant sign determines its real-root count, a classical fact we could not find already in Mathlib.

2. PRELIMINARIES

Throughout, Δ denotes the discriminant of a cubic. For the general cubic $ax^3 + bx^2 + cx + d$,

$$\Delta(a, b, c, d) = 18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2, \quad (1)$$

and classically, for $a \neq 0$: $\Delta > 0$ iff the cubic has three distinct real roots; $\Delta = 0$ iff it has a repeated root (all real); $\Delta < 0$ iff it has one real root and a complex-conjugate pair. For the monic case $a = 1$ we write $\Delta_3(b, c, d)$ for $\Delta(1, b, c, d)$; abusing notation slightly to match the variable names used in Theorems 1–2 (whose monic cubic is $x^3 + ax^2 + bx + c$, i.e. our $(b, c, d) \mapsto (a, b, c)$),

$$\Delta_3(a, b, c) = 18abc - 4a^3c + a^2b^2 - 4b^3 - 27c^2. \quad (2)$$

3. THE MONIC CUBIC

3.1. The band reduction. Write $f(x) = x^3 + ax^2 + bx + c$ and $g(x) = x^3 + ax^2 + bx = f(x) - c$. For $b < a^2/3$, put $s = \sqrt{a^2 - 3b}$; the critical points of f (equivalently of g) are

$$x_- = \frac{-a - s}{3} \text{ (local max),} \quad x_+ = \frac{-a + s}{3} \text{ (local min).}$$

f has three real roots iff $f(x_-) \geq 0 \geq f(x_+)$, i.e. iff c lies in the band

$$c \in [c_{\text{lo}}, c_{\text{hi}}], \quad c_{\text{lo}} = -g(x_-), \quad c_{\text{hi}} = -g(x_+), \quad c_{\text{hi}} - c_{\text{lo}} = \frac{4}{27}s^3.$$

This is not merely an implication but captures the discriminant exactly: as a polynomial in c ,

$$\Delta_3(a, b, c) = -27(c - c_{\text{lo}})(c - c_{\text{hi}}). \quad (3)$$

(3) is verified by direct expansion (and, independently, by `sympy`). For $b > a^2/3$ the right-hand quadratic in c has no real roots and $\Delta_3 < 0$ identically (one real root). Hence for any c -window $[w_{\text{lo}}, w_{\text{hi}}]$,

$$\text{vol} = \iint_{b < a^2/3} \text{len}([c_{\text{lo}}, c_{\text{hi}}] \cap [w_{\text{lo}}, w_{\text{hi}}]) \, db \, da. \quad (4)$$

3.2. Proof of Theorem 1.

Lemma 4 (Never-clipped). *On $D = \{(a, b) : a \in [-1, 1], b \in [-1, a^2/3]\}$, $c_{\text{hi}} \leq 1$, with equality only at $(a, b) = (-1, -1)$.*

Proof. Since $\partial g / \partial a$ at fixed x equals x^2 and $g'(x_+) = 0$ kills the chain-rule term at the critical point, $\partial c_{\text{hi}} / \partial a = -x_+^2 \leq 0$; so c_{hi} is non-increasing in a . As $b \leq a^2/3 \leq 1/3$ for $a \in [-1, 1]$, the point $(-1, b)$ is also in D for every $(a, b) \in D$, so $c_{\text{hi}}(a, b) \leq c_{\text{hi}}(-1, b)$. On the edge $a = -1$: $\partial c_{\text{hi}} / \partial b = -x_+$ with $x_+ = (1 + \sqrt{1 - 3b})/3 > 0$, so c_{hi} is strictly decreasing in b there, with maximum at $b = -1$: $c_{\text{hi}}(-1, -1) = 1$ exactly. $\square$

The map $(a, b, c) \mapsto (-a, b, -c)$ preserves Δ_3 (direct expansion) and the cube, and sends c_{hi} to $-c_{\text{lo}}$; hence $c_{\text{lo}} \geq -1$ on D as well. **The band never exits the window** $[-1, 1]$: it touches the boundary only at the two corner points $(\mp 1, -1, \pm 1)$, of measure zero. Therefore, using (4) with no clipping correction,

$$\text{vol} = \int_{-1}^1 \int_{-1}^{a^2/3} \frac{4}{27} (a^2 - 3b)^{3/2} \, db \, da = \frac{8}{405} \int_{-1}^1 (a^2 + 3)^{5/2} \, da = \frac{3064}{1215} + \frac{8}{3} \operatorname{arsinh} \frac{1}{\sqrt{3}}.$$

Since $\operatorname{arsinh}(1/\sqrt{3}) = \ln(1/\sqrt{3} + 2/\sqrt{3}) = \frac{1}{2} \ln 3$,

$$\mathbb{P} = \text{vol} / 8 = \frac{383}{4860} + \frac{\ln 3}{48}.$$

Remark 5. The corner-touching geometry is exactly why this is solvable in closed form: the naive “main term plus clipping corrections” program has *all* corrections identically zero, so the trivial upper bound already equals the answer. High-precision quadrature landing on that upper bound to sixteen digits was, in practice, the signal that this was happening.

3.3. Proof of Theorem 2. Now the domain is $a \in [0, 1]$, $b \in [0, a^2/3]$, window $[0, 1]$. Since $b \geq 0$ forces $s \leq a$, both critical points satisfy $x_- \leq 0 \leq$ (the argument of the local max) and $x_+ \leq 0$; and $c_{\text{hi}} \geq 0$ because $g(0) = 0$, $g'(0) = b \geq 0$, and $x_+ \leq 0$ is the local minimum, so $g(x_+) \leq 0$. A short computation shows $c_{\text{hi}} \leq 1/27$ throughout the domain, so the top of the window $[0, 1]$ never binds. The bottom does: $\partial c_{\text{lo}}/\partial b = -x_- \geq 0$, and $c_{\text{lo}} = 0$ exactly on the curve $b = a^2/4$ (where $x_- = -a/2$ and $g(-a/2) = 0$); so $c_{\text{lo}} \leq 0$ for $b \leq a^2/4$ and $c_{\text{lo}} \geq 0$ for $a^2/4 \leq b \leq a^2/3$ – a single sign change. Hence the overlap length is c_{hi} on $[0, a^2/4]$ and the full band width on $[a^2/4, a^2/3]$, and, remarkably, all fractional powers cancel:

$$\int_0^{a^2/4} c_{\text{hi}} db + \int_{a^2/4}^{a^2/3} \frac{4}{27} (a^2 - 3b)^{3/2} db = \frac{a^5}{480}.$$

Integrating over $a \in [0, 1]$,

$$\mathbb{P} = \int_0^1 \frac{a^5}{480} da = \frac{1}{2880}.$$

4. THE NON-MONIC CUBIC

Dropping the monic restriction – letting the leading coefficient a range uniformly over $[-1, 1]$, including near 0 – makes the problem genuinely four-dimensional, and Lemma 4’s trick does not generalize directly: it is special to a symmetric window of half-width *exactly* 1 around a *fixed* leading coefficient.

Writing $V(t)$ for the volume of the real-rooted region inside the cube $[-t, t]^3$ in the monic-cubic coefficient space (so $V(1) = 8 \cdot \mathbb{P}$ of Theorem 1), the natural attack conditions on the leading coefficient and rescales:

$$\mathbb{P}(\text{non-monic}) = \int_0^1 \frac{a^3}{8} V(1/a) da. \quad (5)$$

This requires $V(t)$ for every $t \geq 1$, and (5) turns out to be a dead end: high-precision numerics for $p(a) := a^3 V(1/a)/8$ (computed to 40 digits; e.g. $p(1/2) = 0.1732556347784\dots$ and $p(1/3) = 0.2277312312100\dots$) show no recognizable closed form under a blind PSLQ search against $\{1, \ln 2, \ln 3, \ln 5, \sqrt{2}, \sqrt{3}, \pi\}$. $V(1)$ is elementary only because $t = 1$ is exactly where the corner-touching geometry of Lemma 4 applies – not because $V(t)$ is nice in general.

4.1. The discriminant region is a cone. The route that works is structural rather than a direct generalization of the band calculus. The discriminant (1) is homogeneous of degree 4: $\Delta(\lambda x) = \lambda^4 \Delta(x)$ for $x = (a, b, c, d)$, so the real-rooted set $R = \{\Delta > 0\}$ is invariant under $x \mapsto \lambda x$ ($\lambda > 0$) – a cone. Apply the divergence theorem to $F = x/4$ (so $\text{div } F = 1$) on $R \cap [-1, 1]^4$. Euler’s identity gives $x \cdot \nabla \Delta = 4\Delta$, which vanishes identically on the lateral boundary $\{\Delta = 0\}$; so that part of the surface integral is zero, and only the eight faces of the cube contribute:

$$\text{vol}_4(R \cap [-1, 1]^4) = \frac{1}{4} \oint_{\partial(R \cap [-1, 1]^4)} F \cdot n dS = \frac{1}{2} (S_a + S_b + S_c + S_d),$$

where S_a, S_b, S_c, S_d are the (signed, outward) contributions of the four faces $a = \pm 1, \dots$ paired by central symmetry $x \mapsto -x$ (which preserves Δ and the cube). The coefficient reversal $(a, b, c, d) \mapsto (d, c, b, a)$ – sending each root x to its reciprocal $1/x$ – also preserves both R and the cube, and identifies $S_a = S_d$, $S_b = S_c$. Hence

$$\text{vol}_4(R \cap [-1, 1]^4) = V(1) + S_b, \quad \mathbb{P}(\text{non-monic}) = \frac{V(1) + S_b}{16}, \quad (6)$$

where $V(1) = S_a = \frac{766}{1215} + \frac{\ln 3}{6}$ (eight times Theorem 1) and

$$S_b = \text{vol}_3\{(a, c, d) \in [-1, 1]^3 : ax^3 + x^2 + cx + d \text{ has three real roots}\}$$

is the volume on the face where the *second* coefficient is pinned to 1 – a genuinely new, but purely three-dimensional, problem, with neither a $t \rightarrow \infty$ tail nor an $a \rightarrow 0$ endpoint singularity to contend with.

Remark 6. We verified the reduction step (6) itself against a case with an independently known answer: applied to $q^2 > 4pr$ on $[-1, 1]^3$ (also a cone, the quadratic discriminant being homogeneous), the three faces give $S_p = S_r = 13/6$, $S_q = 5/2 + \ln 2$, and $\frac{1}{3} \cdot 2(S_p + S_q + S_r)$ works out to $\frac{41}{9} + \frac{2}{3} \ln 2$ – exactly eight times the known full-quadratic volume $\frac{41}{72} + \frac{\ln 2}{12}$. The reduction is an identity, not an approximation.

4.2. Computing S_b . Substitute $s = \sqrt{1 - 3ac}$ (so $c = (1 - s^2)/(3a)$). The admissible- d band on this face is $[-K_-/u, K_+/u]$ with

$$K_+ = (s - 1)^2(2s + 1), \quad K_- = (s + 1)^2(2s - 1), \quad u = 27a^2, \quad K_+ + K_- = 4s^3,$$

on the domain $\{s \in (0, 2), a \in [|s^2 - 1|/3, 1]\}$. Integrating over a *first*, at fixed s , is elementary, since a enters only through $u = 27a^2$; two lemmas make it precise.

Lemma 7 (Never clips above). $d_{\text{hi}} \leq 1$ throughout: $\alpha_+ := \sqrt{K_+/27} < a_0 := |s^2 - 1|/3$ reduces to $2s + 1 < 3(s + 1)^2$, true for all s . This is the exact analogue of Lemma 4.

Lemma 8 (Clips below exactly for $s \in (2/3, 2)$). $\alpha_- := \sqrt{K_-/27} > a_0$ reduces to $(3s - 2)(s - 2) < 0$; and $\alpha_- < 1$ reduces to $(s + 1)^2(2s - 1) < 27$, i.e. $s < 2$, with equality precisely at $s = 2$.

With Lemmas 7–8, $F(s) := \int_{a_0}^1 L(a, s)/a \, da$ (the a -integral of the band length) is elementary – the term $-K_+/(2K_-) + 2s^3/K_-$ collapses to the constant $1/2$ via $K_+ + K_- = 4s^3$ – giving the piecewise closed form

$$F(s) = \begin{cases} \frac{2s^3}{27} \left(\frac{1}{a_0^2} - 1 \right), & s \leq 2/3, \\ \frac{K_+}{54a_0^2} + \frac{1}{2} + \ln \frac{\alpha_-}{a_0} - \frac{2s^3}{27}, & 2/3 < s < 2, \end{cases}$$

continuous at $s = 2/3$ (both branches equal $11264/18225$ there). Then

$$S_b = \frac{4}{3} \int_0^2 s F(s) \, ds = \frac{1454}{405} - \frac{5}{6} \ln 3 = 2.674613216233365 \dots,$$

and, substituting into (6) with $V(1) = \frac{766}{1215} + \frac{\ln 3}{6}$,

$$\mathbb{P}(\text{non-monic}) = \frac{1}{16} \left(\frac{766}{1215} + \frac{\ln 3}{6} + \frac{1454}{405} - \frac{5}{6} \ln 3 \right) = \frac{641}{2430} - \frac{\ln 3}{24}.$$

(Confirmed to hold exactly by direct symbolic simplification in `sympy`.)

4.3. Independent verification. Two of the checks below are fully independent of each other and of the derivation above; the rest are internal consistency checks of the computation.

- (1) **Exact symbolic evaluation** (as above) – `sympy` confirms $\mathbb{P} - (\frac{641}{2430} - \frac{\ln 3}{24}) = 0$ identically.
- (2) **An independent numerical route, sharing no step with the derivation above:** fixing the leading coefficient a and integrating a out *last* (rather than first, as in (6)) – effectively a direct high-precision evaluation of (5), with the $a \rightarrow 0$ endpoint singularity and the migrating clipping breakpoints of the inner integral carefully handled – reproduces $0.21801049620261477102 \dots$, agreeing with the exact value to *nineteen* significant digits (Gauss–Legendre and tanh–sinh quadrature, several degrees each, spread 1.6×10^{-18}).

- (3) **Blind PSLQ**: given only the 22-digit numeric value of S_b , a PSLQ integer-relation search (no closed form supplied) recovers $S_b = \frac{1454}{405} - \frac{5}{6} \ln 3$ unprompted.
- (4) **Monte Carlo**, 4×10^{10} samples on the raw sign of (1) (using no theory from this paper): $0.218008985 \pm 2.1 \times 10^{-6}$, -0.73σ from the exact value.
- (5) Double-precision brute-force quadrature (no special handling of any kind): 0.2180105004 , 4.2×10^{-9} from the exact value – at that method’s own precision limit.

Remark 9. This integrand carries three numerical hazards worth recording, all concentrated at small leading coefficient a and all invisible to a quadrature routine’s own convergence estimate: (i) the conditional probability given the leading coefficient has a *square-root* endpoint singularity as $a \rightarrow 0$, $p(a) = p(0) - \frac{2}{3}\sqrt{a} + O(a)$ with $p(0) = \frac{1}{2} + \frac{5}{72} + \frac{\ln 2}{12}$ (the random-quadratic limit, since as $a \rightarrow 0$ the third root escapes to $-b/a$), on which a fixed-degree Gauss rule loses roughly seven digits while reporting no warning; (ii) the clipping/kink structure in the inner variable migrates to scale $\sim \sqrt{3a}$ as $a \rightarrow 0$, so any fixed panel layout eventually walks past it; and (iii) floating-point arithmetic loses the clipping breakpoints altogether for small a , since the relevant breakpoint cubic has a near-double root whose branches separate by only $O(\sqrt{a})$. All three had to be identified and handled to reach the nineteen-digit confirmation above.

4.4. Relation to the dxdy.ru thread. The forum thread’s stated intermediate volume, $V_0 = \frac{766}{1215} + \frac{\ln 3}{6}$, agrees with $V(1) = 8 \cdot$ Theorem 1, and its final claimed value, $\frac{641}{2430} - \frac{\ln 3}{24}$, is confirmed exactly by Theorem 3. Its remaining steps therefore do reach the right answer, though nothing in the present paper validates the specific route the thread took to get there; the derivation above is independent.

5. FORMAL VERIFICATION IN LEAN

All three theorems have been formalized and *completely, machine-checked proved* in Lean 4 against Mathlib (toolchain v4.33.0), with **zero** uses of **sorry** anywhere in the development and no unexpected axioms: `#print axioms` on every headline theorem returns exactly `[propext, Classical.choice, Quot.sound]`, the three standard axioms used throughout Mathlib itself. The development is 3780 lines across seven files (`Basic.lean`, `Integrals.lean`, `Theorem1.lean`, `Theorem2.lean`, `Theorem3Statement.lean`, `Theorem3Proof.lean`, `DiscriminantRootCount.lean`), all in the accompanying repository, and rebuilds from a warm Mathlib cache in under a minute.

The discriminant–root-count bridge. Everywhere above, “has three real roots” is proved as a statement about the *sign of* Δ . Mathlib has `Cubic.discr` and a criterion for `discr  $\neq 0$` implying distinct roots over a splitting field, but nothing connecting the *sign* of a real cubic’s discriminant to how many of its roots are real – the fact stated without proof in Section 2 above, and standard since at least Cardano. `DiscriminantRootCount.lean` proves it from scratch and with no appeal to calculus: for $p = b^2 - 3ac > 0$, evaluating $27a^2f$ at the two points where $3at + b = \mp\sqrt{p}$ gives values whose *product* is $-27a^2\Delta$ and whose *difference* is $4p^{3/2}$, and three applications of the intermediate value theorem to the resulting sign pattern produce three real roots directly, with no derivative or critical point appearing anywhere. Theorems 1 and 2 are additionally restated and proved as `theorem1_root_count` and `theorem2_root_count`, in terms of `HasThreeDistinctRealRoots` directly, with no discriminant in the statement at all.

Two presentational simplifications found while formalizing. The informal proof of Theorems 1–2 above derives the band via the critical points $x_{\pm}$ of g . In Lean, it proved cheaper to skip $x_{\pm}$ entirely and use the single ring-provable identity

$$-27 \Delta_3(a, b, c) = (27c - 9ab + 2a^3)^2 - 4(a^2 - 3b)^3,$$

which yields both branches of (3) at once, and to prove Lemma 4 from the closed forms $27c_{hi} = (a - s)^2(a + 2s)$, $27c_{lo} = (a + s)^2(a - 2s)$ directly, reducing the whole lemma to a single scalar inequality on $a \in [-1, 1]$, $s \in [0, 2]$ that `nlinarith` discharges automatically with product hints.

More strikingly, the cone step of §4 ((6)) needs *no divergence theorem at all*. The Lean proof of `volume_conePiece1` establishes, for any measurable cone $R \subseteq \mathbb{R}^4$, that the piece of $[-1, 1]^4$ where coordinate 1 attains $\max_j |x_j|$ has volume $\frac{1}{4} \text{vol}_3(\text{face})$, by slicing at $x_1 = t$, using homogeneity of R to identify the slice with $t \cdot (\text{face})$, and integrating $\int_0^1 t^3 dt = \frac{1}{4}$. This is Fubini plus the elementary scaling law $\text{vol}_3(t \cdot E) = t^3 \text{vol}_3(E)$: no boundary integral appears anywhere, so the cusps of $\{\Delta = 0\}$ never have to be dealt with, and no smoothness of ∂R is used or needed. The seven remaining pieces follow from the first by measure-preserving coordinate maps built from the two `ring`-provable symmetries of Δ . Both simplifications are the same mathematics as the informal derivation, just in a form that turned out to need less machinery than either the critical-point picture or the classical divergence theorem statement suggested – and neither changes any of the numbers above.

Remark 10. Formalizing surfaced one genuine error and confirms none of the paper’s numerical claims were wrong. An early Lean statement of the inner integral $F(s)$ ((5)’s companion, defined for $s \in (0, 2)$) was false exactly at $s = 1$, where $a_0 = 0$ and the integral genuinely diverges; both sides collapsed to unrelated junk values under Lean’s total-function conventions ($0/0 = 0$, $\log 0 = 0$) rather than raising an error, and only attempting the proof – not merely writing the statement – caught it. The fix is an $s \neq 1$ hypothesis, harmless since $\{1\}$ is null. We record this because it is a direct illustration of why every numerical claim in this paper rests on multiple independent computations rather than on any one of them: an unchecked claim, even one that type-checks, is not yet a fact.

6. NOVELTY AND PRIOR ART

An eight-agent parallel literature sweep (arXiv, journal databases, Math StackExchange and MathOverflow’s own search indices, OEIS, and general web search), followed by a targeted search on the exact constants derived here, found:

- **Theorem 1:** no prior statement of this probability anywhere searchable (OEIS direct digit-sequence query: no match; Math StackExchange/MathOverflow site search for the monic cubic: zero threads; web search for the exact fraction and decimal: empty). Two caveats. First, the *unnormalized* volume $V_0 = \frac{766}{1215} + \frac{\ln 3}{6}$ (i.e. eight times our value) appears, uninterpreted as a probability, as an intermediate step of the `dx dy.ru` thread discussed above – an unrefereed, partly AI-assisted forum derivation of the *non-monic* cubic, dated June 2026, not indexed by any search engine at the time of our search. Second, Li [1] claims an exact expression for the cubic family $x^3 + 3ax^2 + 3bx + 2c$ with (a, b, c) uniform on symmetric intervals $[-h, h] \times [-k, k] \times [-l, l]$; at $h = k = 1/3$, $l = 1/2$ this specializes to exactly our model. The paper itself is paywalled and, per its recovered *Mathematical Reviews* notice [2], no numerical or symbolic value from it has ever been reproduced by either of its two citations; whether it correctly reduces to Theorem 1 cannot be checked without the original text.
- **Theorem 2:** likely new without caveat – Li’s furnished special cases are all symmetric intervals, and the value does not appear in the `dx dy.ru` thread or anywhere else searched.
- **Theorem 3:** the *value* is not new (see above); the *proof* and its independent verification are.

We regard the Li [1] question as the one genuinely open item in this account.

7. OPEN PROBLEMS

The reduction techniques used here appear to generalize.

- (1) **Monic quartic.** The quartic discriminant is homogeneous of degree 6 in the *five* coefficients a, b, c, d, e of $ax^4 + bx^3 + cx^2 + dx + e$, so the divergence-theorem argument of §4 does apply – but to the *non-monic* quartic on $[-1, 1]^5$, reducing that all-real-roots volume to a sum over $2 \times 5 = 10$ three-dimensional face volumes. One of those faces (the one with the leading coefficient pinned to 1) is exactly the *monic* quartic probability we would actually like; the reduction therefore needs it as an *input*, not a consequence, and does not apply directly to the monic problem stated here. Fixing the leading coefficient at 1 breaks the homogeneity the cone argument relies on, so the monic quartic likely needs a direct generalization of the band argument of §3 instead – complicated by a quartic’s derivative being itself a cubic, with up to three real critical points rather than one band. Current Monte Carlo estimate: $0.0054749 \pm 9.0 \times 10^{-6}$; under active investigation at the time of writing.
- (2) **Gaussian coefficients.** For the monic cubic with (a, b, c) i.i.d. standard normal, there is no cube to bound the band, so Lemma 4 does not apply in any form; the natural object is $\mathbb{E}[\Phi(c_{\text{hi}}) - \Phi(c_{\text{lo}})]$, a smooth two-dimensional integral with no case splits, but a priori no reason to expect an elementary closed form. Current Monte Carlo estimate: $0.169962 \pm 4.2 \times 10^{-5}$; under active investigation at the time of writing.
- (3) **General degree n .** The probability of no real roots for a degree- n Kac polynomial decays as $n^{-3/4}$ exactly (Poplavskiy–Schehr [7]); the corresponding rate constant for *all* n roots real, under speed- n^2 large-deviation scaling, is not known in closed form for uniform (or Kac-Gaussian) coefficients.
- (4) **The parent matrix problem.** For an $n \times n$ matrix with i.i.d. uniform entries, the all-real-eigenvalue probability remains open for $n \geq 3$; the striking empirical near-coincidence $\mathbb{P}_n(\text{uniform}) \approx 2^{-(n-1)(n-2)/4}$ (Edelman’s Gaussian probability at rank $n-1$), first observed numerically on Math StackExchange, has no known explanation.

APPENDIX A. NUMERICAL VERIFICATION SUMMARY

	Theorem 1	Theorem 2	Theorem 3
Symbolic (<code>sympy</code> , exact)	PASS	PASS	PASS ($\mathbb{P} - \text{cand.} = 0$)
Independent numerical route	$\sim 10^{-16}$	$\sim 10^{-13}$	19 sig. digits (7.0×10^{-20})
Blind PSLQ / <code>identify</code>	—	—	recovers exact form
Monte Carlo	2×10^8 : $+1.3\sigma$	2×10^8 : -0.5σ	4×10^{10} : -0.73σ
Lean 4 / Mathlib	0 sorry	0 sorry	0 sorry

DATA AND CODE AVAILABILITY

All source code (Python: `sympy/mpmath/numpy/scipy`; Lean 4: Mathlib v4.33.0), raw numerical results, the literature-sweep records underlying §6, and an interactive visual explainer of the monic-cubic case are available in the accompanying repository.

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